GenderBender P23 — Manual

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What GenderBender P23 is

GenderBender P23 is a MIDI drum-pattern generator for REAPER that plays the four voices of the SOMA Pulsar-23: **BD**, **BASS**, **SD** and **HHT**. It is a JSFX plugin built on GenderBender (original version by Sergio Millán, GPL v3), and it sends notes only — the sound comes from the Pulsar.

Each channel picks a **style**, a set of 9 patterns of up to 32 steps. An **XY pad** blends between them, and **DENS** takes notes away or adds them one by one. Swing, slide and breaks shape the groove, and the plugin schedules every note to the exact sample.

The library has 193 styles: the 31 originals plus 162 new ones from Galicia, world music, electronic, hip hop, industrial, experimental and ambient. Styles can be in 4/4, 3/4, 6/8, 5/8 or 7/8.

Pulsar-23 channel	Plays from the style	MIDI out
BD	kick	channel 10, note 36
BASS	toms as a bass line (low tom = note, high tom = note + ALT +)	channel 2, note 36
SD	snare + clap	channel 10, note 38
HHT	closed hat + open hat	channel 10, note 42

Installation

Unzip `GenderBenderP23_reaper.zip` over REAPER's resource folder (in REAPER: *Options → Show REAPER resource path*). The zip mirrors that folder:

Path in the zip	What it is
`Effects/GenderBenderP23/genderbender_p23.jsfx`	the plugin

Path in the zip	What it is
`Data/drum_styles_v/*.txt`	the 193 styles
`Data/drum_styles_v_list.txt`	backup list of style names, used if REAPER cannot list the folder

From the repository, `sh tools/install.sh` does the same and rebuilds the list. If you add styles by hand and the header says *list file*, run the installer again or add the file name to the list.

Then insert **MIDI GenderBender P23** on a track. If you update the plugin, remove it from the track and insert it again so it reads the parameters fresh.

Setting up the Pulsar-23

The plugin follows REAPER's tempo and transport; the Pulsar only needs MIDI in on its DIN socket.

1. **Route the MIDI.** In the track's routing window, set the hardware MIDI output to the interface connected to the Pulsar.
2. **Learn each module once.** Hold the module's **LRN** button on the Pulsar (BD, SD, HHT or BASS) and click **TEST** on the matching channel in the plugin. The Pulsar keeps the assignment after power-off. Its MIDI LED turns green for assigned notes and red for notes it does not know.
3. **BASS mode.** Set the BASS switch to **MIDI** to hear the line with its octave jumps (ALT +). In **CV** mode it responds to the learned key only, so set ALT + to 0. In **PRC** mode, SOMA's manual says it follows pitch when its channel differs from the drums; check this on your unit.
4. **Shape the sound on the Pulsar.** GATE is how long each note holds the Pulsar's envelope open; ATT and REL on each module still shape it. Gates of a few hundred ms turn BD, SD or HHT into sustained tones or noise.
5. **Clock (optional).** The plugin does not need the Pulsar in sync. To run the Pulsar's own dividers and loop recorders at tempo too, enable *Send clock/SPP* for that MIDI output in REAPER, set the Pulsar's clock switch to **MIDI** and patch **LRST** to **0.25**, as its manual recommends.

Channels and notes are fixed: BD, SD and HHT on channel 10 (36, 38, 42) and BASS on channel 2 (36). The BASS needs its own channel because in MIDI mode it plays every note that reaches its channel.

The interface

The window has a header, four channel columns and a global panel on the right. It scales with the window size.



Area	What it holds
Header	Plugin name, the number of styles found (and *list file* if they came from the backup list), and three buttons: **ALL RND** (one random style for all four channels), **EACH RND** (a random style per channel) and **SRC RST** (sources back to KICK, TOMS, SN+CLAP, HH CL+OP).
Channel column	Name and activity LED (flashes on every note, brighter for higher velocity), TEST button, STYLE (with the meter shown when it is not 4/4), SOURCE, the step view, and the knobs. The MIDI output of the channel is shown at the bottom.
Step view	The pattern that is playing now, for the current pad position, density and bar. Colour = the step plays; white = an ALT hit (clap, open hat, high tom); short colour = a fill on an empty step; faint bar = a step in the pattern that density leaves out; white mark below = playhead. Steps beyond the loop of a 3/4, 6/8, 5/8 or 7/8 style are shown dark.

Area	What it holds
Right panel	XY pad, global SWING, and the BREAK controls (EVERY, LENGTH, OFFSET, RANDOM).

Using the knobs. Drag up or down to change a value; hold Shift for fine steps; or use the mouse wheel. Right-click or Ctrl+click returns a knob to its default. Every parameter can be automated from REAPER.

Channel parameters

All four channels have the same controls; only ALT + is BASS-only.

Control	Range (default)	What it does
STYLE	any style in the list	The style file this channel reads. Arrows step through the list, the wheel scrolls, a click opens the full menu. The name is white when loaded, grey while it loads and red if the file could not be opened.
SOURCE	10 options (BD: KICK · BASS: TOMS · SD: SN+CLAP · HHT: HH CL+OP)	Which of the style's 7 voices the channel plays. Single voices: KICK, SNARE, CLAP, HH CL, HH OP, HI TOM, LO TOM. Pairs: SN+CLAP, HH CL+OP and TOMS, where the second voice is the ALT voice with its own gate. When both voices hit the same step, the ALT voice wins.
DENS	0–100 % in 0.1 steps (25 %)	How many notes play. 25 % = the pattern as written. Below, notes are removed one by one; above, ornaments, borrowed hits and fills are added one by one. See the next section.
DIV	x1/8 … x4 (x1)	Speed of the step grid. x1 plays the pattern as written (32 steps per 4/4 bar, 32nd notes); x2 and x4 play it 2 or 4 times faster; x1/2 … x1/8 stretch it over 2 to 8 bars; x1.5 and x3 are triplet grids that drift against the bar.

Control	Range (default)	What it does
BREAK	OFF, or a grid like DIV (OFF)	The grid the channel switches to inside the break window set in the right panel, for rolls and stutters. OFF = this channel ignores breaks.
VEL	1–127 (127)	Maximum MIDI velocity.
DYN	0–100 % (50 %)	How much the step's level sets the velocity: velocity = VEL × ((1 − DYN) + DYN × level/255). At 0 % every note is at VEL; at 100 % ornaments become ghost notes.
GATE	1–2000 ms (BD 30 · BASS 150 · SD 30 · HHT 15)	Note length, which on the Pulsar is how long the envelope stays open (sustain).
ALT GT / OPEN	1–2000 ms (BD 30 · BASS 150 · SD 30 · HHT 250)	Gate of ALT hits. On HHT it is labelled OPEN: a long gate is the open hat, and the next closed hat cuts it.
SWING	0–100 % (100 %)	How much of the global swing this channel uses: effective swing = 50 + (global − 50) × amount.
SLIDE	−30 to +30 ms (0)	Moves the whole channel earlier (negative, pushes) or later (positive, drags). The plugin works 30 ms ahead, so early notes are never lost.
ALT +	−24 to +24 semitones (+12), BASS only	Added to the BASS note on ALT hits: with TOMS, the high tom plays an octave up. Needs the BASS in MIDI mode.
TEST	button	Sends the channel's note now, even with the transport stopped, for the Pulsar's LRN learn.

How DENS works

DENS counts notes rather than comparing levels, so every part of its travel changes something. At 0 % nothing plays, at 25 % the channel plays exactly the hits marked X in the pattern, and at 100 % it plays every step of the loop.

Each step gets a priority, and notes enter in this order:

1. **X** — the pattern's essential hits. Between 0 and 25 % they come in one by one, strongest positions first.
2. **x, o, -** — the style's ornaments, from strongest to weakest.
3. **Borrowed hits** — steps where this voice plays in any of the style's other 8 patterns. They keep the added notes in the style's language.
4. **Fills** — the remaining empty steps, up to all of them at 100 %. Only the main voice fills; the ALT voice stops at its borrowed hits.

Inside each group, the beats of the bar come first, then 8ths, 16ths and 32nds, following the channel's meter (in 6/8 the dotted-quarter beats lead). Ties are broken by a fixed pseudo-random order.

Variation over time. The X hits always keep the same order, so the skeleton of the rhythm stays put. Everything DENS adds above 25 % changes order every two bars over an 8-bar cycle, and each new note enters one bar of the cycle at a time. The result: hundreds of distinct knob positions (165–200 measured on typical styles, against 25–35 before), and a groove that evolves instead of repeating the same loop. At 0, 25 and 100 % all bars are the same.

Velocity of added notes. With DYN above 0, notes that DENS adds play softer than the pattern's own (borrowed hits at 60 % of their level, fills at a low level), with a small fixed variation so they do not all sound alike.

In breaks the channel uses 75 % of its DENS, minus a random amount of up to RANDOM points.

Global parameters

XY pad

The pad blends the 9 patterns of every channel's style at once. Drag the dot; right-click or Ctrl+click returns it to the centre. X and Y run from −100 to +100. Each label is the pattern pair that sounds at that point (p4 = p4 in odd bars and p5 in even bars; p8 has no pair):

	Left (X −100)	Centre (X 0)	Right (X +100)
Top (Y +100)	p0	p8	p6
Middle (Y 0)	p4	p4	p2
Bottom (Y −100)	p0	p0	p8

The styles follow one layout: **p4/p5 BASE** in the centre, **p2/p3 ENERGY** to the right, **p6/p7 VARIATION** top right, **p0/p1 INTRO** left and bottom, **p8 FILL** top centre and bottom right. Inside each quadrant the four corners are mixed smoothly; crossing an axis changes quadrant and can jump. This is the original GenderBender formula, kept so every style behaves the same in both plugins.

SWING

50–75 % (50 %), MPC-style: the second 16th of each 8th moves later, to 0.125 beats at 75 %. Each channel takes its share with its own SWING knob.

BREAK

A break window repeats every few beats. Inside it, each channel with BREAK on switches to its break grid and plays at 75 % of its DENS minus up to RANDOM points.

Control	Range (default)	Meaning
EVERY	0–32 beats (4)	Length of the cycle; 0 = no breaks.
LENGTH	1–32 beats (1)	How many beats at the end of the cycle the break lasts.
OFFSET	0–32 (4)	The break never starts before this beat of the cycle.
RANDOM	0–100 % (0 %)	Random density taken away in the break, new on every step.

With the defaults, the last beat of every 4 is a break.

Meters

A style can be in 4/4, 3/4, 6/8, 5/8 or 7/8; the tag in its name tells which, and the channel loops with that length. The meter also shows next to STYLE, and the step view darkens the steps outside the loop.

Tag in the name	Meter	Loop	Strong positions	Examples
none	4/4	32 steps (4 quarter notes)	every quarter	most styles

Tag in the name	Meter	Loop	Strong positions	Examples
(3-4)	3/4	24 steps (3 quarters)	every quarter	XOTA 168, FOLIADA 150, VALSE 160
(6-8)	6/8	24 steps (6 eighths)	the two dotted quarters	MUINEIRA NOVA 180, RIBEIRANA 186, TARANTELLA 190
(5-8)	5/8	20 steps	2 + 3	PAIDUSHKO 160
(7-8)	7/8	28 steps	2 + 2 + 3	RUCENICA 140, KALAMATIANOS 130, SNARECORE 170

Set REAPER to the same meter so bars line up with the loop. The tempo in a style's name is in quarter notes: a muiñeira at 180 is a dotted quarter of 120.

Polymeter. Each channel loops at its own length, so different meters on different channels phase against each other — for example BD in 4/4 against HHT in 7/8.

The original GenderBender always plays 32 steps, so meter styles sound cut short there.

The style library

The 193 styles live in `Data/drum_styles_v/` and are shared with the original GenderBender. `docs/fingerprints.html` in the repository shows every pattern of every style, grouped by family.

Family	Styles	Examples
Galicia	24	Muiñeira nova and vella, ribeirana, carballesa, xota, foliada, alborada, pasodobre, mazurca, valse, muiñeira techno
Latin America & Caribbean	19	Salsa, guaguancó, samba, maracatu, candombe, reggae one drop, dembow
Iberia & Mediterranean	14	Sevillanas, rumba flamenca, tarantella, maqsum, gnawa, rucenica (7/8)
IDM & experimental	13	Braindance, algorítmica, polyrhythms, Euclidean, *Clapping Music*
House & disco	12	Deep, acid and garage house, disco, italo, UK funky

Family	Styles	Examples
Africa	11	Afrobeat, highlife, amapiano, kuduro, singeli
Hip hop	11	Boom bap, Dilla swing, lo-fi, trap, UK drill
Club & bass	10	Miami bass, Jersey club, ballroom, witch house
Asia & Middle East	5	Dangdut, keherwa and teentaal (tabla), dadra (6/8), bhangra
Techno, trance, breaks & DnB	23	Detroit, dub, acid, psytrance, gabber, amen jungle, neurofunk
Trip hop, industrial, breakcore, ambient	20	Bristol, EBM, rhythmic noise, breakcore, ambient dub
Original GenderBender styles	31	Cumbias, juke, house, grime, dubstep, gqom

Galician styles name their voices in Galician: kick = bombo, snare = tamboril/caixa, clap = pandeireta skin or shells, closed hat = ferreñas, open hat = jingle roll, toms = bass/drone. The meter, accent and character of each Galician form come from published descriptions; the exact bombo, tamboril and pandeireta parts are written for this library.

Pattern layout and levels

Each style holds 7 voices × 9 patterns × 32 steps, each step 0–255. Levels: **X** = 255 (plays at DENS 25 %), **x** = 170, **o** = 120, **-** = 60 (added by DENS), **.** = 0.

Writing your own style

Write a `.style` file in `styles_src/` and build it:

```
name: MY STYLE 120
meter: 6/8          # optional; adds "(6-8)" to the name
[p4 p5]             # the lines below go into these patterns
kick   X.....X.....
hhc    X.x.x.X.x.x.
```

Then run `python3 tools/make_style.py build styles_src/my_style_120.style` and `python3 tools/check_styles.py` (it must print `ok`). In 4/4 a line has 16 characters (16ths) or 32 (32nds); in 3/4 and 6/8, 12 or 24; in 5/8, 10 or 20; in 7/8, 14 or 28. Voices: `kick`, `snare`, `clap`, `hhc`, `hho`, `hitom`, `lotom`. Never edit the `.txt` files by hand: a broken file silences the channel.

Troubleshooting

Symptom	What to check
Header says *no styles found*	The styles are not in `Data/drum_styles_v/` of the REAPER resource folder, or `drum_styles_v_list.txt` is missing from `Data/`.
Style name stays grey	It loads on the next audio block; press play or arm the track.
Style name turns red	The file could not be opened or is broken: rebuild it with `make_style.py`.
A Pulsar module does not play	Repeat the LRN + TEST learn for that module. The Pulsar's MIDI LED is red when it receives a note it has not learned.
The BASS plays drum notes	The BASS learned channel 10. Learn it again with the BASS TEST button (channel 2).
High tom (ALT) hits are silent on the BASS	The BASS is not in MIDI mode; switch it, or set ALT + to 0.
A 6/8 or 3/4 style does not line up with bars	Set REAPER's project time signature to the style's meter.
Nothing plays	The track's MIDI hardware output is not set, the transport is stopped, or DENS is at 0 %.

Quick reference

- **25 % DENS** = the pattern as written; 0 % = silence; 100 % = every step.
- **Pad centre** = BASE (p4/p5); right = ENERGY; top right = VARIATION; left/bottom = INTRO; p8 = FILL.
- **Right-click** any knob or the pad = default value.
- **Out:** BD ch 10 note 36 · SD ch 10 note 38 · HHT ch 10 note 42 · BASS ch 2 note 36.
- **Licence:** GPL v3, as GenderBender (original version by Sergio Millán).